SmartGrow OS — Project Summary

Stack: React + Vite + Tailwind CSS v4 (frontend), Node.js + Express + MongoDB/Mongoose (backend), JWT authentication
Repository: github.com/Iresha2002/greenhouse-system- (branch `main`)

The 4 Modules

#	Module	What it does
1	Smart Greenhouse	Live sensor monitoring, climate zones and actuator control (already existed).
2	Crop Marketplace	Farmers sell fresh harvests. Customers get a cart, checkout and order history; farmers have a fulfilment portal (already existed).
3	Agricultural Suppliers Marketplace	Built in an earlier session (11 stages).
4	ExpertConnect	Replaced the old Hydroponic Bay. Built in 12 stages.

Suppliers Marketplace (Stages 1–11)

- **Catalog:** 10 product categories; search by chemical formula (NPK, EC, pH) or protocol (RS485, LoRaWAN).
- **Customer side:** server-verified basket, checkout (cash on delivery or warehouse pickup), order tracking and cancellation with stock restored, profile page.
- **Supplier portal:** 8 KPI cards, product editor, inline stock controls, order workflow (Pending to Delivered).
- **Rankings:** best sellers computed from real orders, with no fake ratings.

ExpertConnect (Stages 1–12)

Stage	What was built
1–2	Landing page; navigation renamed from Hydroponic Bay.
3	`expert` role, ExpertProfile model, directory with search and filters, profile cards and modal, 4 seeded experts.
4	Expert application form and admin approve/reject. Nobody can verify themselves.
5–6	Problem reports with up to 5 photos (max 2 MB each) and rule-based expert matching that explains each suggestion.
7	Booking with live slots (Sri Lanka time), expert availability editor, double-booking blocked (409).
8	Private chat per consultation, with photos and highlighted diagnostic notes.

9	Structured expert recommendation (observations, causes, actions, safety, follow-up) with an advice disclaimer.
10	In-app notifications; 1–5 star reviews only after a completed consultation, updating the expert's rating.
11	Admin moderation tab: analytics, all consultations, review removal, account suspension.
12	Full 8-workflow test run and README documentation.

Security Built In

- Role-based access for farmer, expert, supplier, customer and admin.
- Farmers cannot see each other's consultations; experts can only act on their own.
- Suspended accounts are blocked at login and on every request.
- Image uploads validated for type and size.

Test Accounts

Role	Email	Password
Admin	admin@smartgrow.lk	admin123
Experts	kavinda.expert@, tharushi.expert@, roshan.expert@, nadeesha.expert@smartgrow.lk	expert123
Farmers / suppliers	supplier@smartgrow.lk, nutrients@ceylonagritech.lk	supplier123